

Program : Diploma in Instrumentation Engineering	
Course Code : 2081	Course Title: Electronic Instrumentation
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the performance characteristics of instruments.
- To learn the basic concepts of Electrical and Electronic Instruments such as Moving coil and moving iron instruments, digital multimeters.
- To learn the basic concepts.
- To familiarize the use of DC and AC bridges.
- To familiarize the calibration of watt meter and energy meter.
- To familiarize the working of CRO/DSO, LED, LCD, strip chart and X-Y recorder.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic knowledge of electronic components		Fundamentals of Electrical & and electronics	Secondary School
Fundamentals of electric and magnetic circuits		Fundamentals of Electrical & and electronics	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the performance characteristics of instruments.	12	Applying

CO2	Utilize moving coil meter as multimeter.	9	Applying
CO3	Explain the construction and working of watt meters, energy meters and AC & DC Bridges.	11	Understanding
CO4	Explain the working of display devices –CRO, DSO, LED & LCD, Strip chart recorders and XY recorders.	11	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the performance characteristics of instruments.		
M1.01	Explain measurements and instrumentation system- Generalized block diagram and different system of units.	4	Understanding
M1.02	Identify performance characteristics of instruments- static and dynamic, Different types of errors, calibration.	8	Applying

Contents:

Introduction to Instrumentation Engineering: Generalized block diagram of instrumentation system; Different system of Units, conversion between them; Performance characteristics of instruments- Static characteristics (accuracy, precision, linearity, sensitivity, resolution, error, hysteresis, range or span, repeatability, reproducibility, drift) Dynamic characteristics- (dynamic error, fidelity, measuring lag, speed of response); Different types of error - gross error, systematic error and random error.

CO2	Utilize moving coil meter as multimeter.		
M2.01	Explain working of moving coil and moving iron instruments and construction of PMMC galvanometer.	3	Understanding
M2.02	Utilize the PMMC as voltmeter and ammeter and multimeter.	3	Applying
M2.03	Explain digital multimeter with block diagram.	3	Understanding
	Series Test – I	1	
Contents: Module-2: Moving Coil Instruments, Moving Iron Instruments and Multimeters: Working of moving coil instruments and moving iron instruments. Construction and working of PMMC galvanometer with basic equation. Conversion of PMMC in to voltmeter and ammeter and multimeter. Difference between analog and electronic multimeter. Working of digital multimeter with block diagram.			
CO3	Explain the construction and working of watt meters, energy meters and AC & DC Bridges.		
M3.01	Construct different AC and DC Bridges.	8	Applying
M3.02	Explain working of Watt meters and Energy meters	3	Understanding
Contents: Bridges, Watt meter and Energy meter: DC Bridge- Wheatstone bridge – derivation of balance equation; AC Bridges- Derivation of balance equation, Inductance measurement using Maxwell bridge and Hay bridge- balance equation; Capacitance measurement using Schering bridge, Wien bridge- balance equation; Construction and working of dynamometer type watt meter; Construction and working of induction type single phase energy meter.			
CO4	Explain the working of display devices –CRO, DSO, LED & LCD, Strip chart recorders and XY recorders.		
M4.01	Summarize the working of CRO with block diagram	2	Understanding
M4.02	Explain the working of DSO with block diagram,	2	Understanding

M4.03	Explain the working of LED and LCD.	3	Understanding
M4.04	Explain the working of Strip chart recorder and XY recorder.	4	Understanding
	Series Test – II	1	
Contents: Module-4- CRO, DSO, Display Devices and Recorders: Working of CRO with block diagram; Working of DSO with block diagram; Working of LED displays- (seven segment and dot matrix) and LCD; Working of Strip chart recorder (potentiometric type) and XY recorder.			

Text / Reference

T/R	Book Title / Author
R1	B L Thereja, “Electrical Technology”, S Chand.
R2	AK Sawhney, “A Course in Electrical and Electronic Measurements and Instrumentation”, Dhanpath Rai & Sons.
R3	Kalsi, “ Electronic Instrumentation”, TMH.
R4	William D Cooper, “Modern Electronic Instrumentation and Measurement Techniques”, PHI.
R5	David A. Bell, “Electronic Instrumentation and Measurements”, Second Edition, Oxford University Press.

Online Resources

Sl.No	Website Link
1	www.ieee-ims.org
2	www.nptel.ac.in/courses
3	http://vlab.co.in/
4	https://www.electronicsworld.com/
5	http://www.instrumentationtoolbox.com/